

# Alvaro Guerra

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## EDUCATION

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### Texas A&M University

*B.S. Mechanical Engineering*

College Station, TX

2021 - 2025

## EXPERIENCE

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### Structures Lead

*IGNITORS (Rocketry Team)*

11-2024 – 06-2025

College Station, TX

- Owned airframe design of 11k-ft Hybrid COTS rocket, led hazardous parachute pyrotechnic testing, ensuring effectiveness of ejection charges in conjunction with altimeter electronics.
- Managed \$3.5k BOM and vendor contracts, delivering on schedule and 1% under budget.

### Structures and Material Science Designer

*SAE Aero Design Team*

05-2022 – 04-2023

College Station, TX

- Established test criteria and pass/fail requirements for multiple flight tests with varying amounts of hardware payload.
- Achieved 1st place internationally among 40+ by troubleshooting and repairing critical systems minutes before the final competition flight.

### Garden Robot Developer

*Garden Cultivation Capstone Project*

08-2024 – 05-2025

College Station, TX

- Designed fluid and electrical control systems for remote control irrigation, finishing 100% of design goals, all \$200 under budget.
- Designed and built Arduino/Pi-driven assistive garden robot electronic schematics; wrote embedded C++ to control drives, pumps, and sensors, boosting water efficiency 26%

### Real Estate Analyst Intern

*River Valley Appraisals of Texas, LLC*

05-2021 – 08-2021

Edinburg, TX

- Built regression models to refine comparable sales market analysis with Excel on 100+ properties

## RELEVANT COURSEWORK

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### MEEN 345 Fluid Mechanics Laboratory (CFD, DAQ)

11-2023

- Strengthened systems-level thinking by bridging theoretical fluid dynamics with hands-on sensor-based testing and simulation-driven design feedback
- Validated aerodynamic performance via hands-on testing in the AF1300 subsonic wind tunnel; integrated VDAS software to simulate airflow, define boundary conditions, and collect drag performance data

### MEEN 423 Machine Learning Engineering Data Analysis (Python)

01-2021 – 04-2021

- Applied ML methods to stress-strain and financial datasets to automate anomaly detection—strengthened data acquisition and interpretation skills essential for integrated test systems.

## TECHNICAL SKILLS

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**Software:** SolidWorks, Python (*pandas*, *NumPy*, *Matplotlib*, *BeautifulSoup*, *PyQt*), MATLAB, HTML/CSS, Git, Microsoft Office, LabVIEW, VDAS

**Testing & Firmware:** Sensors (NEO6M, DHT22, NI MyDAQ, Arduino, Raspberry Pi, Wind Tunnel (AF1300), Oscilloscope, Multimeter

**Additional:** English / Spanish fluency, TRA L1 / L2 (in progress)